Unit 6, Lesson 19

Divide with Remainders

Standards

Building On **4.OA.A.3, 4.OA.B.4**

Addressing **4.NBT.B, 4.NBT.B.6, 4.OA.A.3, 4.OA.B.4**

Instructional Routines

- Notice and Wonder

Goals

- Comprehend (in spoken and written language) the meaning of the term "remainder."
- Explain (orally) strategies for dividing up to four-digit numbers by one-digit divisors.
- Interpret (orally) the quotient and remainder for a situation involving division.

Student Facing Learning Goals

Let's solve problems involving division including interpreting remainders.

Lesson Purpose

The purpose of this lesson is to introduce division expressions that have remainders.

Narrative

Students have divided four-digit divisors by one-digit dividends using place value understanding and the properties of operations. In this lesson, students work with division situations and expressions that don't have whole-number quotients. Students connect **remainders** to the idea of "leftovers" when dividing. They conceptualize remainders by determining that an additional equal group cannot be made with the remaining part of the dividend, or by determining that the dividend is not a multiple of the divisor.

In these materials, division that results in a whole number with a remainder—for example, $145 \div 7$—is not expressed with an expression such as "20 R 5." Instead, students will relate this result to a multiplication equation, $145 = 7 \times 20 + 5$.

Access for Students with Disabilities

- Engagement

Access for English Learners

- MLR8

Lesson Timeline

Warm-up	10 min
Activity 1	25 min
Activity 2	10 min
Synthesis Estimate	10 min
Cool-down	5 min

Teacher Reflection Questions

What evidence of multiplication fluency can be collected from student thinking in this lesson? How might you use this evidence to support development of fluency for students who need support?

Warm-up

10 min

Notice and Wonder: Expressions and Equations

Standards

Addressing **4.NBT.B, 4.OA.B.4**

Instructional Routines

- Notice and Wonder

This *Warm-up* prompts students to analyze patterns and look for structure in division expressions and equations (MP7), and to reinforce their understanding of factors and multiples. This *Notice and Wonder* prepares students to work with division situations that have remainders later in the lesson.

Student Task Statement

What do you notice? What do you wonder?

division expression	multiplication equations
$100 \div 5$	$100 = 20 \times 5$
$101 \div 5$	$101 = 20 \times 5 + 1$
$102 \div 5$	$102 = 20 \times 5 + 2$
$103 \div 5$	$103 = 20 \times 5 + 3$
$104 \div 5$	$104 = 20 \times 5 + 4$
$105 \div 5$	$105 = 20 \times 5 + 5$

Student Response

Students may notice:

- All the division expressions have a dividend of 5.
- 100 and 105 are the only dividends that are multiples of 5.
- Each time the dividend goes up one, you add one more in the equation.
- In each row, the total in the equation is the same as the dividend.

Students may wonder:

- Why do some equations have addition and some don't?
- Why is it adding more each time?
- What do the equations have to do with the expressions?

Launch

- Groups of 2
- Display the table.
- "What do you notice? What do you wonder?"
- 1 minute: quiet think time

Activity

- "Discuss your thinking with your partner."
- 1 minute: partner discussion
- Share and record responses.

Activity Synthesis

- "Today, we are going to work with dividing numbers that don't have a whole number quotient."

Activity 1

$182 \div 6$

25 min

Standards

Building On **4.OA.A.3**

Addressing **4.NBT.B.6**

The purpose of this activity is to introduce students to division situations that don't have whole number quotients. Students write division situations to represent an expression. Once students generate a list of situations, students ask a question and solve one of the problems that does not involve dividing the remainders into pieces. Students create a poster of their work and do a Gallery Walk, with a focus on how each group represented the leftovers in the situation. At the end of the *Activity Synthesis* the term **remainder** is introduced to describe "leftovers" when dividing.

MLR8 Discussion Supports. To support the transfer of new vocabulary to long-term memory, invite students to chorally repeat phrases that include the word "remainder" in unison, 1–2 times.
Advances: Speaking

Teacher Display

Student Task Statement

$182 \div 6$

Write division situations that the expression can represent.

Student Response

Sample responses:

- 182 students go on a field trip in which they are put into groups of 6.
- There are 182 students going on a field trip and 6 buses.
- There are 182 books to be shared equally between 6 classes.
- 182 cans of food need to be packed in 6 boxes.

Launch

- Groups of 2
- "Take a few minutes with your partner to come up with as many division situations as you can that the expression could represent."
- 3–4 minutes: partner work time

Activity

- As students share, record responses on a chart. Be sure to record just the situations without the questions.
- Choose a situation in which the context does not support dividing the leftovers into pieces.
- "What question could we ask about this situation?"
- Record questions.
- "Work with your partner to answer the question. Explain or show your reasoning."
- "Create a visual display that shows your thinking about the problems. You may want to include details such as notes, diagrams, drawings, and so on, to help others understand your thinking."
- 6–8 minutes: independent or group work
- 5–7 minutes: gallery walk

Activity Synthesis

- "What is the same and what is different about the way students represent the leftovers?"
- 30 seconds quiet think time
- 1 minute: partner discussion
- "This problem had a **remainder**. When we divide and end up with leftovers we call them remainders, because they represent what is remaining after we divide into equal groups."

Activity 2

10 min

Is There a Remainder?

Standards

Building On **4.OA.B.4**

Addressing **4.NBT.B.6, 4.OA.A.3**

The purpose of this activity is for students to use what they know about factors and multiples to determine if division expressions will have remainders. Students analyze expressions without finding the value and then choose to find the value of one expression that will result in a remainder and one without a remainder.

Engagement: Develop Effort and Persistence. Differentiate the degree of difficulty or complexity. Begin with an expression with more accessible values. For example, $22 \div 5$.
Supports accessibility for: Social-Emotional Functioning, Conceptual Processing

Student Task Statement

1. Decide if each expression will result in a remainder. Explain how you know.
   a. $753 \div 6$
   b. $918 \div 9$
   c. $1{,}263 \div 2$
   d. $2{,}630 \div 5$
2. Find the value of 2 expressions. Choose one with a remainder and one without. Explain or show your reasoning.

Student Response

1. a. It will have a remainder. Sample response: Since 753 isn't even, 6 cannot be a factor.
   b. It will not have a remainder. Sample response:

Launch

- Groups of 2
- "We are going to figure out if division expressions will result in a remainder. Look at the first problem and be prepared to share your reasoning with your partner."
- 2 minutes: quiet think time
- 2 minutes: partner discussion

Activity

- Monitor for students who use numerical reasoning to identify whether the expressions will have remainders.
- Invite selected students to share their reasoning for the first problem.
- "Now, find the value of 2 expressions, one with a

900 is a multiple of 9 and 18 is a multiple of 9.

c. It will have a remainder. Sample response: Since 1,263 is odd, it is not a multiple of 2.

d. It will not have a remainder. Sample response: Since 2,630 ends in 0, it is a multiple of 5.

2. Sample responses:

a. 125 with a remainder of 3. Partial quotients notation with partial quotients of 600, 120 and 30.

b. 102. $900 \div 9 = 100$ and $18 \div 9 = 2$ and $100 + 2 = 102$

c. 631 with a remainder of 1. Partial quotients of 600, 30 and 1.

d. 526. Partial quotients of 2,000, 500, 100 and 30.

remainder and one without.

- 3–4 minutes: independent work time

Activity Synthesis

"In the *Lesson Synthesis*, let's represent quotients that have leftovers when we take away as many equal groups as we can from a number."

Lesson Synthesis

"Today, we saw that sometimes when you divide numbers, you get a remainder."

Display $753 = 6 \times 125 + 3$.

"How does this equation show that $753 \div 6$ has a remainder?" (It shows 753 is not a multiple of 6. It also shows 6 and 125 make a factor pair for 750, and 753 is 3 more than that.)

"How can we write an equation to show that $1{,}263 \div 2$ has a remainder?" ($1{,}263 = 2 \times 631 + 1$)

Cool-down

5 min

Remainders

Standards

Addressing **4.NBT.B.6**

Student Task Statement

Find the value of $1{,}865 \div 4$. How many are left over?

Use partial quotients to show your reasoning.

Student Response

466 groups with a remainder of 1. Sample response:

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   466
     6
    10
    50
   400
8 )1,865
- 1,600
   265
-  200
    65
-   40
    25
-   24
     1
```

Responding to Student Thinking

Students use many partial quotients and lose track of the answer.

Next Day Supports
Launch *Warm-up* or the first activity by highlighting important notation from previous lessons.